MATH 241: Calculus III

Fall 2026

Lecture 16: Triple Integrals

Background & Motivation

Everything we did with double integrals extends naturally to three dimensions. Triple integrals compute volume, mass, and moments over solid regions. Cylindrical and spherical coordinates play the same role that polar coordinates did for double integrals—they simplify regions with the right symmetry.

1. Warm-Up: Double Integral Review

(a) Evaluate $\int_0^2 \int_1^3 xy\,dy\,dx$.

1. Inner (y): ________________________________
2. Outer (x): ________________________________
3. Result: ________________

(b) Evaluate $\iint_R y\,dA$ where R is bounded by $y = 0$, $x = 0$, and $x + y = 1$.

1. Region (Type I): $0 \le x \le 1$, $0 \le y \le 1 - x$.
2. Inner: ________________________________
3. Outer: ________________________________

(c) Evaluate $\int_0^{2\pi} \int_0^1 r^2 \cdot r\,dr\,d\theta$.

1. Inner (r): ________________________________
2. Outer (θ): ________________________________

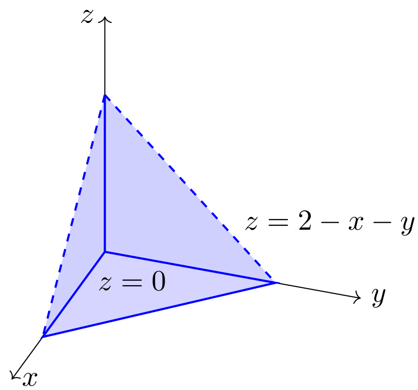
2. Triple Integrals and Fubini's Theorem

Definition 1: Triple Integral

$$\iiint_E f(x, y, z)\,dV = \int\int\int f(x, y, z)\,dz\,dy\,dx$$

– When $f = 1$: the triple integral gives the **volume** of E

– Fubini's theorem: for continuous f on a box $[a, b] \times [c, d] \times [p, q]$, integrate in any order



A typical triple-integral Type I region: the solid tetrahedron in the first octant bounded below by $z = 0$ and above by the plane $z = 2 - x - y$. Integrate z first from the bottom to top surface, then project onto the xy-plane for the remaining double integral.

Types of Regions

- **Type I:** z bounded by functions of (x, y) – project onto the xy-plane
- **Type II:** x bounded by functions of (y, z) – project onto the yz-plane
- **Type III:** y bounded by functions of (x, z) – project onto the xz-plane

Choose the type that makes the bounds simplest.

3. Examples in Rectangular Coordinates

Example 1: Rectangular Box

Evaluate $\int_0^1 \int_0^2 \int_0^3 (x + y + z)\, dz\, dy\, dx$.

Example 2: Type I: First-Octant Tetrahedron

Find the volume of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

4. Cylindrical Coordinates

Cylindrical Coordinates

$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$

$$dV = r\,dz\,dr\,d\theta$$

– Use when the region has circular symmetry in x, y (cylinders, cones, paraboloids)
– Same as polar coordinates in xy, with z added

Example 3: Cone Volume via Cylindrical

Find the volume of the solid bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by $z = 3$.

5. Spherical Coordinates

Spherical Coordinates

$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$

$$dV = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$$

– ρ: distance from origin $\quad$ ϕ: angle from positive z-axis ($0 \le \phi \le \pi$) $\quad$ θ: angle in xy-plane
– Use when the region is a sphere/hemisphere or integrand involves $x^2 + y^2 + z^2$
– Key relation: $x^2 + y^2 + z^2 = \rho^2$

Example 4: Integral over a Sphere

Evaluate $\iiint_E (x^2 + y^2 + z^2)\,dV$ where E is the ball $x^2 + y^2 + z^2 \le 4$.

Practice Problems

Problem 1. Evaluate $\int_0^1 \int_0^1 \int_0^1 (x + y + z)\,dz\,dy\,dx$.

Problem 2. Evaluate $\iiint_E z\,dV$ where E is bounded below by $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$. (Use cylindrical.)

Problem 3. Find the volume of the upper hemisphere $x^2+y^2+z^2 \leq 9$, $z \geq 0$ using spherical coordinates.

Problem 4. Evaluate $\iiint_E \sqrt{x^2+y^2+z^2}\, e^{-(x^2+y^2+z^2)}\, dV$ over all of $\mathbb{R}^3$. (Use spherical.)